Theorems and Laws for Boolean Algebra

Boolean Algebra Operators	OR: +	AND: .	NOT: '
	XOR: ⊕	XNOR: ≡ (aka: Equivalence)	

Expression Substitution	XYZ, or any other expression, may be substituted with one variable, A	
Additive Identities	$A + 0 = A$	$A + 1 = 1$
Multiplicative Identities	$0. A = 0$	$1. A = A$
Idempotent Law	$A + A = A$	$A. A = A$
Laws of Complementarity	$A + A' = 1$	$A. A' = 0$
Commutative Law	$X. Y = Y. X$	$X + Y = Y + X$
Associative Law	$(X + Y) + Z = X + (Y + Z) = X + Y + Z$	$(XY). Z = X. (YZ) = XYZ$
Distributive Law	$X(Y + Z) = XY + XZ$	$X + YZ = (X + Y)(X + Z)$
Simplification Theorem	$XY + XY' = X$	$(X + Y)(X + Y') = X$
	$X + XY = X$	$X(X + Y) = X$
	$(X + Y')Y = XY$	$XY' + Y = X + Y$
DeMorgan's Laws	$(A + B)' = A'B'$	$(AB)' = A' + B'$
Duality	$(XYZ)^D = X + Y + Z$	$(X + Y + Z)^D = XYZ$
XOR and Equivalence Operations	$X \oplus Y = X'Y + XY'$	$X \equiv Y = X'Y' + XY$
Consensus Theorem	$XY + X'Z + YZ = XY + X'Z$ (Notice that YZ disappeared)	